Supplement — Paired-Acquisition Neural Factorization: An End-to-End Computational Pathology Pipeline

Public preprint supplement — 2026-09-04. Research use only; not clinical software.

1 Implementation-path inventory

Table 1 lists the primary implementation paths per research line, bound in the evidence manifest (`evidence/manuscript_evidence_manifest.json`). Every architectural claim in the main text binds to these source paths and their tests.

Table 1: Implementation-path inventory.

Line	Primary implementation paths
PCam	`experiments/train_pcam.py`; `evaluate_pcam.py`; `src/data/datasets/pcam_dataset.py`
PA-NF	`experiments/paired_acquisition/` runners; `src/paired_acquisition_factorial.py`
SCORPION	`experiments/paired_acquisition/run_*scorpion*`; `src/models/scorpion_pathoalign.py`
Canine	`run_biological_label_preservation_fixed_estimand.py`
TransnnMIL	`src/models/transnnmil/`; `src/models/mil/`
PathologyFL	`src/features/federated/pathology_fl/`
FAIR-WEIGHTS-H	`.../weighting/fair_weights_h.py`; `.../pathoalign_bridge.py`
CAMELYON17	`experiments/camelyon17_*`; `scripts/camelyon17/`
Reproducibility	`scripts/provenance/`; `src/paired_acquisition_provenance.py`

2 Test coverage

Table 2 maps each line to its test suites. Tests are the binding target for architectural claims.

Table 2: Test coverage per research line.

Line	Test suites
PCam	`tests/test_pcam_*.py`; `test_pcam_evaluation_ci.py`
Paired/SCORPION/canine	`tests/test_fixed_estimand_real_feature_space_adjudication*.py`; `tests/test_real_paired_scanner_*.py`
Synthetic	`tests/test_biological_bottleneck_capacity_allocation_factorial.py`
TransnnMIL	`tests/models/test_transnnmil_*.py`; `test_topology_branch.py`; `test_hierarchical_pooling.py`
PathologyFL	`tests/federated/test_fl_integration.py`; `test_pathology_fl_*.py`; `test_secure_aggregation*.py`
FAIR-WEIGHTS-H	`tests/federated/test_fair_weights_h.py`; `test_weighting_*.py`
CAMELYON17	`tests/test_camelyon_*.py`; `tests/camelyon17/`
Reproducibility	`tests/test_corrected_paired_acquisition_evidence.py`; `test_real_bottleneck_representation_recovery.py`

3 Result artifact hashes

The authoritative artifact hashes are recorded in `evidence/manuscript_evidence_manifest.json`. Selected current hashes:

- corrected paired-acquisition release manifest: `evidence/paired_acquisition/corrected-20260726/release_manifest.json`;
- SCORPION capacity-matched release manifest: `evidence/paired_acquisition/scorpion-capacity-matched-20260726/release_manifest.json`;
- dimensionality/cross-covariance release manifest: `evidence/paired_acquisition/dimensionality-xcov-factorial-20260726/release_manifest.json`.

4 PA-NF controlled-result boundary

The primary registered SCORPION comparison is the 175-fit capacity-matched campaign: seven variants, five folds, and five seeds. PA-NF reduces tissue-branch scanner balanced accuracy by 0.3108 relative to the equal-capacity two-branch neural control while satisfying the registered same-region retrieval noninferiority criterion and retaining strong acquisition-branch scanner information. The corrected canine SCC comparison answers a broader and harder question. Under the fixed five-category estimand, strong centroid/QR and paired-linear corrections remain competitive and no universal neural feature-space increment is established. The two results use different comparators and endpoints and should be interpreted together rather than collapsed into one ranking.

5 Whole-slide and institutional result pointers

The full PANDA feature manifest contains 10,611 readable Phikon slide bags. TransnnMIL optimization-stability results are recorded under `docs/results/panda-transnnmil-stability.md`. The full-cache dominance-aware detector experiment is recorded under `docs/results/dominance-aware-switch-full-panda.md`, and the fixed label-noise-to-ordinal-shift transfer is recorded under `docs/research/dominance-detector-transfer-results.md`.

CAMELYON17 source-weighting and center-subspace experiments are recorded under `results/camelyon17/`, `results/camelyon17_supervised_resnet18/`, and the corresponding research notes.

6 Reproducibility

The repository preserves code, tests, result summaries, release manifests, and source hashes for the major research lines. Detailed correction history and superseded analyses remain available in the repository but are supporting research records rather than the narrative focus of the flagship manuscript.